

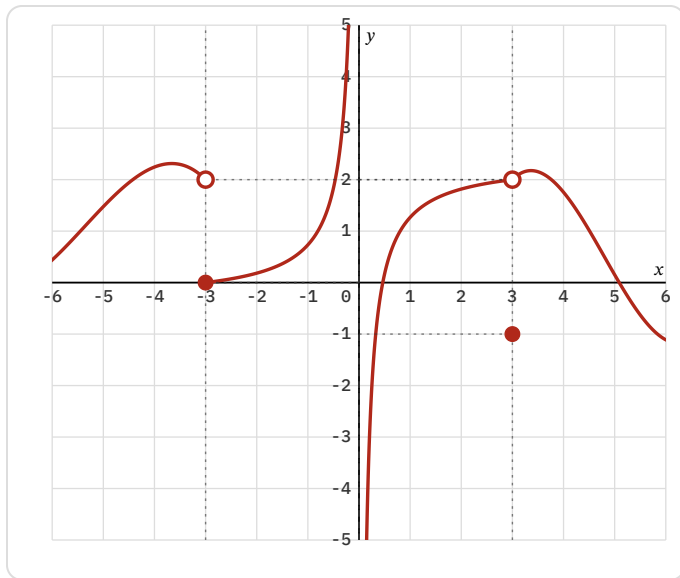
Mock Midterm C — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 88 points

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write **DNE**.



a.) $f(0)$ (1)

DNE

b.) $\lim_{x \rightarrow 0^-} f(x)$ (1)

∞

c.) $\lim_{x \rightarrow 0^+} f(x)$ (1)

$-\infty$

d.) $\lim_{x \rightarrow 0} f(x)$ (1)

DNE

e.) $f(3)$ (1)

-1

f.) $\lim_{x \rightarrow 3^-} f(x)$ (1)

2

g.) $\lim_{x \rightarrow 3^+} f(x)$ (1)

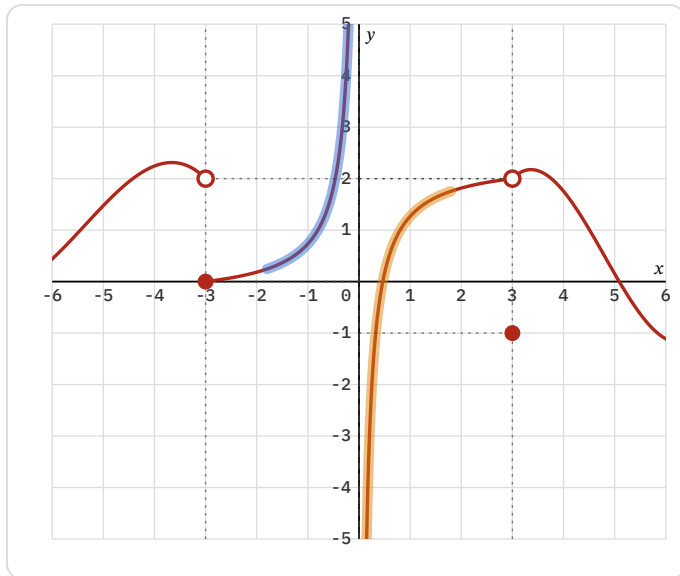
2

h.) $\lim_{x \rightarrow 3} f(x)$ (1)

2

8 pts

WORKED SOLUTION



— approaching $x = 0$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

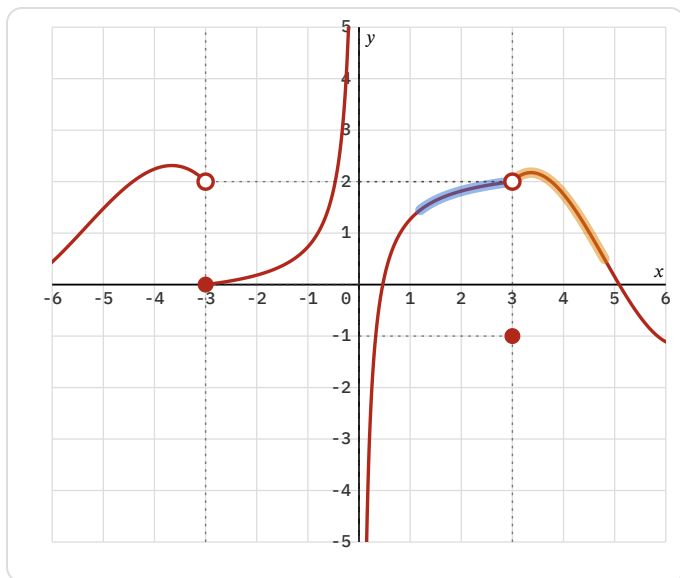
At $x = 0$

There is no filled dot on the line $x = 0$, so $f(0)$ is undefined $\rightarrow$ DNE.

From the left, the curve runs along the asymptote $x = 0$ upward, so the limit is ∞ .

From the right, the curve runs along the asymptote $x = 0$ downward, so the limit is $-\infty$.

The left and right limits don't match, so $\lim_{x \rightarrow 0} f(x)$ DNE.



— approaching $x = 3$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

At $x = 3$

The **filled** dot at $x = 3$ sits at height -1, so $f(3) = -1$.

Tracing the curve toward $x = 3$ from the left, the heights approach 2 (open or closed dot doesn't matter).

Tracing the curve toward $x = 3$ from the right, the heights approach 2 (open or closed dot doesn't matter).

Both one-sided limits equal 2, so $\lim_{x \rightarrow 3} f(x) = 2$ — even though $f(3) = -1$. The limit ignores the value AT the point.

-
2. Let $f(x) = \frac{1}{(x+2)^2}$. Answer with a number, ∞ , $-\infty$, or DNE. 5 pts
- a.) $\lim_{x \rightarrow -2^-} f(x)$ (1)
 ∞
- b.) $\lim_{x \rightarrow -2^+} f(x)$ (1)
 ∞
- c.) $\lim_{x \rightarrow -2} f(x)$ (1)
 ∞
- d.) $\lim_{x \rightarrow \infty} f(x)$ (1)
 0
- e.) The vertical asymptote is $x =$ (1)
 -2
-

WORKED SOLUTION

Near $x = -2$ the denominator $(x + 2)^2$ is tiny, so $f(x)$ is huge in size. Only the sign needs work.

Squaring makes the denominator a small **positive** number on both sides, so both one-sided limits are ∞ . Since both sides agree we write $\lim_{x \rightarrow -2} f(x) = \infty$ (it still has no finite value).

As $x \rightarrow \infty$, the denominator grows without bound while the top stays 1, so $f(x) \rightarrow 0$. The vertical asymptote is $x = -2$.

3.

Suppose

4 pts

$$\lim_{x \rightarrow 3} f(x) = 4, \quad \lim_{x \rightarrow 3} g(x) = 2, \quad \lim_{x \rightarrow 3} h(x) = 9$$

Use the properties of limits to find each limit.

a.) $\lim_{x \rightarrow 3} [3f(x) + g(x)]^2$ (1)

196

b.) $\lim_{x \rightarrow 3} ([f(x)]^3 - g(x))$ (1)

62

c.) $\lim_{x \rightarrow 3} \sqrt{h(x)} f(x)$ (1)

12

d.) $\lim_{x \rightarrow 3} [3f(x) - 4g(x)]$ (1)

4

WORKED SOLUTION

a. $\lim_{x \rightarrow 3} [3f(x) + g(x)]^2 = [3(4) + (2)]^2 = (14)^2 = 196$

b. $\lim_{x \rightarrow 3} ([f(x)]^3 - g(x)) = (4)^3 - (2) = 64 - 2 = 62$

c. $\lim_{x \rightarrow 3} \sqrt{h(x)} f(x) = \sqrt{9} \cdot (4) = 3 \cdot (4) = 12$

d. $\lim_{x \rightarrow 3} [3f(x) - 4g(x)] = 3(4) - 4(2) = 4$

The quotient property works here because the bottom limit isn't 0.

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{x \rightarrow -3} \frac{3x^2 + 6x - 9}{3x^2 + 14x + 15}$ (2)

3

b.) $\lim_{x \rightarrow 9} \frac{x - 9}{\sqrt{x} - 3}$ (2)

6

4 pts

WORKED SOLUTION

a.

$$\lim_{x \rightarrow -3} \frac{3x^2 + 6x - 9}{3x^2 + 14x + 15} = \lim_{x \rightarrow -3} \frac{(x + 3)(3x - 3)}{(x + 3)(3x + 5)} = \lim_{x \rightarrow -3} \frac{3x - 3}{3x + 5} = \frac{-12}{-4} = 3$$

b.

$$\lim_{x \rightarrow 9} \frac{x - 9}{\sqrt{x} - 3} = \lim_{x \rightarrow 9} \frac{(\sqrt{x} - 3)(\sqrt{x} + 3)}{\sqrt{x} - 3} = \lim_{x \rightarrow 9} (\sqrt{x} + 3) = 6$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use $\infty / -\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow -\infty} \frac{5}{-3x^3 + 4x^2}$ (2)
 0

b.) $\lim_{x \rightarrow \infty} \frac{-2x^3 - 4x^2}{-x + 6}$ (2)
 ∞

c.) $\lim_{x \rightarrow -\infty} \frac{3x + 2}{4x + 9}$ (2)
 $\frac{3}{4}$

WORKED SOLUTION

a. Divide every term by x^3 , the highest power of x in the denominator:

$$\lim_{x \rightarrow -\infty} \frac{5}{-3x^3 + 4x^2} = \lim_{x \rightarrow -\infty} \frac{\frac{5}{x^3}}{-3 + \frac{4}{x}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow -3$, so the limit is 0 .

b. Divide every term by x , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{-2x^3 - 4x^2}{-x + 6} = \lim_{x \rightarrow \infty} \frac{-2x^2 - 4x}{-1 + \frac{6}{x}}$$

The bottom $\rightarrow -1$, but the top is dominated by $-2x^2$, which $\rightarrow -\infty$ as $x \rightarrow \infty$; dividing by -1 gives ∞ .

c. Divide every term by x , the highest power of x in the denominator:

$$\lim_{x \rightarrow -\infty} \frac{3x + 2}{4x + 9} = \lim_{x \rightarrow -\infty} \frac{3 + \frac{2}{x}}{4 + \frac{9}{x}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{3}{4} = \frac{3}{4}$.

6. Find each limit (end behavior). Use ∞ or $-\infty$ when the outputs grow without bound. 4 pts

a.) $\lim_{x \rightarrow \infty} (3x^5 + 3x^4 - 3x^2 - 3)$ (1)
 ∞

b.) $\lim_{x \rightarrow \infty} (4 - 4e^x)$ (1)
 $-\infty$

c.) $\lim_{x \rightarrow -\infty} (-1 + \frac{1}{x^2})$ (1)
 -1

d.) $\lim_{x \rightarrow \infty} (-x^5 + x^2 - 6x)$ (1)
 $-\infty$

WORKED SOLUTION

a. Leading term $3x^5$: degree 5 (odd), leading coefficient positive. As $x \rightarrow \infty$, x^5 is positive, times 3 $\rightarrow \infty$.

b. As $x \rightarrow \infty$, e^x grows without bound, and multiplying by -4 gives $-\infty$.

c. $\frac{1}{x^2} \rightarrow 0$ as $x \rightarrow -\infty$, leaving -1.

d. Leading term $-x^5$: degree 5 (odd), leading coefficient negative. As $x \rightarrow \infty$, x^5 is positive, times -1 $\rightarrow -\infty$.

7. Let

6 pts

$$f(x) = \begin{cases} \frac{x^2 + x - 6}{x + 3} & x < -3 \\ 2x + 5 & x > -3 \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 1} f(x)$ (1)

7

b.) $\lim_{x \rightarrow -6} f(x)$ (1)

-8

c.) $f(-3)$ (1)

DNE

d.) $\lim_{x \rightarrow -3^-} f(x)$ (1)

-5

e.) $\lim_{x \rightarrow -3^+} f(x)$ (1)

-1

f.) $\lim_{x \rightarrow -3} f(x)$ (1)

DNE

WORKED SOLUTION

For $x < -3$, factor the top: $\frac{(x+3)(x-2)}{x+3} = x - 2$ (valid since $x \neq -3$ there).

a. $1 > -3 \rightarrow$ bottom piece: $2(1) + 5 = 7$.

b. $-6 < -3 \rightarrow$ top piece, simplified: $-6 - 2 = -8$.

c. Neither piece includes $x = -3$, so $f(-3)$ is undefined $\rightarrow$ DNE.

d. Left: $\lim_{x \rightarrow -3^-} (x - 2) = -5$.

e. Right: $\lim_{x \rightarrow -3^+} (2x + 5) = -1$.

f. $-5 \neq -1$, so the limit DNE.

8. Find the value of k that makes f continuous at $x = 2$.

4 pts

$$f(x) = \begin{cases} x^2 - 2x & x < 2 \\ kx - 1 & x \geq 2 \end{cases}$$

a.) $\lim_{x \rightarrow 2^-} f(x)$ (2)

0

b.) $k =$ (2)

$\frac{1}{2}$

WORKED SOLUTION

Left: $(2)^2 - 2(2) = 0$. Right (and $f(2)$): $k(2) - 1$.

$$2k - 1 = 0 \Rightarrow k = \frac{1}{2} = \frac{1}{2}$$

-
9. Classify each function at the given x -value.

3 pts

a.) $f(x) = \begin{cases} x & x < 3 \\ x - 2 & x \geq 3 \end{cases}$ at $x = 3$ (1)

(C) Jump discontinuity

b.) $f(x) = \frac{x-3}{x^2-2x-3}$ at $x = -1$ (1)

(D) Infinite discontinuity

c.) $f(x) = \frac{x+1}{x^2-x-2}$ at $x = -1$ (1)

(B) Removable discontinuity

WORKED SOLUTION

a. Left $\rightarrow 3$, right $\rightarrow 1$. Different finite values $\rightarrow$ jump.

b. Factor: $\frac{(x-3)}{(x-3)(x+1)}$. The $(x+1)$ factor does NOT cancel, so $x = -1$ is a vertical asymptote $\rightarrow$ infinite.

c. Factor: $\frac{(x+1)}{(x+1)(x-2)}$. The $(x+1)$ cancels, leaving $\frac{1}{x-2} \rightarrow -\frac{1}{3} \rightarrow$ removable.

10.

Let $C(q) = 2q^2 + 14q + 67$, where $C(q)$ represents the cost, in dollars, of producing q items. Find the average rate of change over $[1, 3]$.

a.) $C(1)$ (2)

83

b.) $C(3)$ (2)

127

c.) Average rate of change (number only) (2)

22

d.) Which sentence correctly states the result **with units**? (1)

(D) Between $q = 1$ and $q = 3$ items, the cost changes at an average rate of 22 dollars per item.

7 pts

WORKED SOLUTION

$$C(1) = 83, \quad C(3) = 127$$

$$\frac{C(3) - C(1)}{3 - 1} = \frac{127 - (83)}{2} = \frac{44}{2} = 22 \text{ dollars per item}$$

It's an *average* over an interval (not an instant), and the units are dollars per item — dollars per item. The total change would be 44 dollars.

11. The table shows the average price of a gallon of regular gas.

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Price, \$	2.37	2.37	2.22	2.10	1.98	1.92	2.08	2.27	2.21	2.45	2.57

Rounded answers (2+ decimal places) are accepted.

- a.) Average rate of change from 2016 to 2020 (2)

$$-\frac{39}{400}$$

- b.) Average rate of change from 2023 to 2026 (2)

$$\frac{1}{10}$$

- c.) The units of these answers are... (1)

(A) dollars per year

5 pts

WORKED SOLUTION

2016–2020:

$$\frac{1.98 - 2.37}{2020 - 2016} = \frac{-0.39}{4} \approx -0.098$$

2023–2026:

$$\frac{2.57 - 2.27}{2026 - 2023} = \frac{0.3}{3} \approx 0.1$$

Units: output units per input unit → **dollars per year.**

12.

Let $f(x) = x^3 - 2x$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) (3)

$$3x^2 + 3xh + h^2 - 2$$

b.) $x = 1, h = 1$ (1)

$$5$$

c.) $x = 1, h = 0.1$ (1)

$$\frac{131}{100}$$

d.) $x = 1, h = 0.01$ (1)

$$\frac{10301}{10000}$$

e.) As $h \rightarrow 0$, what value do these approach? (1)

$$1$$

7 pts

WORKED SOLUTION

$$f(x+h) = (x+h)^3 - 2(x+h) = x^3 + 3x^2h + 3xh^2 + h^3 - 2x - 2h$$

$$\frac{f(x+h) - f(x)}{h} = \frac{3x^2h + 3xh^2 + h^3 - 2h}{h} = 3x^2 + 3xh + h^2 - 2$$

Now substitute:

$$x = 1, h = 1 : 5$$

$$x = 1, h = 0.1 : \frac{131}{100} = 1.31$$

$$x = 1, h = 0.01 : \frac{10301}{10000} = 1.0301$$

Letting $h \rightarrow 0$ leaves 1: the instantaneous rate of change (slope of the tangent line) at $x = 1$. That's exactly what section 1.4 formalizes.

13.

Let $f(x) = \sqrt{x}$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ (3)
 $\frac{1}{2\sqrt{x}}$

b.) $f'(4) =$ (2)
 $\frac{1}{4}$

c.) Slope of the tangent line to f at $x = 9$ (2)
 $\frac{1}{6}$

d.) What does $f'(4) = \frac{1}{4}$ mean? (1)
(B) The instantaneous rate of change of f at $x = 4$ is $\frac{1}{4}$ — the slope of the tangent line there.

8 pts

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}} \end{aligned}$$

Then $f'(4) = \frac{1}{4}$ and the slope of the tangent at $x = 9$ is $f'(9) = \frac{1}{6}$.

$f'(4)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14. (a) Suppose $f(1) = -5$ and $f'(1) = -5$. Write the equation of the tangent line to f at $x = 1$.
(b) The tangent line to the graph of g at $x = 2$ is $y = -3x + 3$. Find $g(2)$ and $g'(2)$.
- a.) (a) Tangent line (3)
 $y = -5x$
- b.) (b) $g(2)$ (2)
 -3
- c.) (b) $g'(2)$ (2)
 -3

7 pts

WORKED SOLUTION

(a) Point $(1, -5)$, slope -5:

$$y + 5 = -5(x - 1) \Rightarrow y = -5x$$

(b) The tangent line touches g at $x = 2$, so $g(2)$ is the line's height there:
 $-3(2) + 3 = -3$. Its slope is the derivative: $g'(2) = -3$.

15.

Let $T(t) = -t^2 + 30t + 70$ be the temperature, in °F, of an oven t minutes after it is turned on.

a.) $T'(t) =$ (3)
 $-2t + 30$

b.) $T'(10)$ (2)
 10

c.) Interpret $T'(10) = 10$. (1)

(C) At $t = 10$ minutes, the oven's temperature is increasing at a rate of 10 degrees Fahrenheit per minute.

6 pts

WORKED SOLUTION

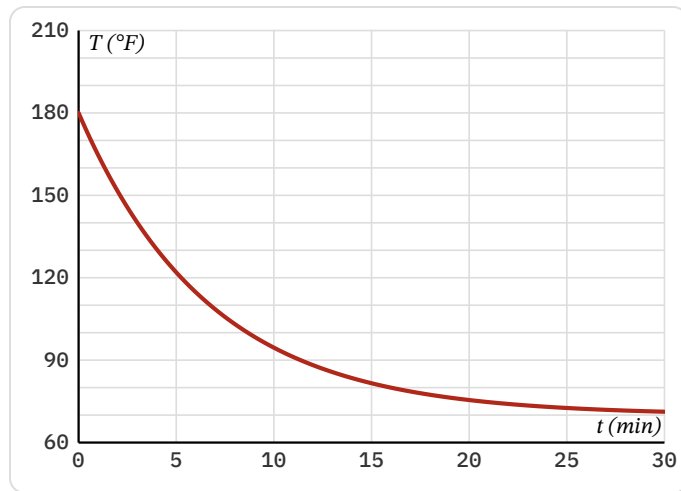
$$\begin{aligned} T'(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{[-(t+h)^2 + 30(t+h) + 70] - (-t^2 + 30t + 70)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-t^2 - 2th - h^2 + 30t + 30h + 70 + t^2 - 30t - 70}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2th - h^2 + 30h}{h} = \lim_{h \rightarrow 0} \frac{h(-2t - h + 30)}{h} \\ &= \lim_{h \rightarrow 0} (-2t - h + 30) = -2t + 30 \end{aligned}$$

$$T'(10) = -2(10) + 30 = 10$$

Positive/negative tells the direction, and the units are degrees Fahrenheit per minute: at that instant the oven's temperature is increasing at 10 degrees Fahrenheit per minute. (The value $T(10) = 270$ is a different quantity.)

16.

A cup of coffee is left on a counter. Its temperature T (in $^{\circ}\text{F}$) after t minutes is graphed below. Decide whether each statement is true or false.



- a.) The instantaneous rate of change at $t = 25$ is closer to 0 than the average rate of change over $[0, 10]$. (1)

True

- b.) The coffee is cooling faster at $t = 3$ minutes than at $t = 20$ minutes. (1)

True

- c.) The tangent lines get steeper as t increases. (1)

False

- d.) The average rate of change over $[0, 10]$ is positive. (1)

False

4 pts

WORKED SOLUTION

a. **True.** $T'(25) \approx -0.39$ $^{\circ}\text{F}/\text{min}$ vs. $\text{ARC} \approx -8.5$ $^{\circ}\text{F}/\text{min}$.

b. **True.** The graph is steeper at $t = 3$ (slope ≈ -10.5 $^{\circ}\text{F}/\text{min}$) than at $t = 20$ (≈ -0.8 $^{\circ}\text{F}/\text{min}$).

c. **False.** The curve levels off toward 70 $^{\circ}\text{F}$, so the tangent lines flatten (slopes move toward 0).

d. **False.** The temperature drops from 180 to about 94.5, so the $\text{ARC} \approx -8.5$ $^{\circ}\text{F}/\text{min}$ — negative.
